



भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

CS6903: Network Security

(M0: Introduction and Logistics)

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CSE Department

IIT Hyderabad

Acknowledgement

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 - Prof. Bheemarjuna Reddy Tamma, Prof. Kotaro Kataoka, Prof. Antony Franklin
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 - Sandeep K. Shukla, Stefan Savage, Dan Boneh, Kirill Levchenko, Alex Gantman, Deian Stefan, Nadia Heninger, Alex Dent, Vitaly Shamtikov, Robert Turner
- ❑ Other internet sources

Course Instructors



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Teaching Assistance (TBA)

- Sakshi Srivastava

- A few more...

Why Should We Learn Network Security?

❑ The Growing Need

- **Digital Dependency:** Business, governments, and individuals rely heavily on interconnected systems
- **Cyber Threats:** Ransomware, phishing, DDoS attacks are increasing in frequency and sophistications
- **Skills Gap:** Cybersecurity vacancies are projected to exceed 3.5 million by 2025¹

1. <https://cybersecurityventures.com/jobs-report-2021/>

Why Should We Learn Network Security?

❑ Benefit of Expertise

- **Career Opportunities:** High-paying roles like Network Security Engineers, Penetration Tester, and Security Architect
- **Critical Role:** Protecting sensitive data and ensuring uptime for organizations
- **Personal Safety:** Safeguard your devices, network and online identities

The Problem of Cyber Threats

❑ Real-world Impact

- **Financial Loss:** Cybercrime costs are projected to reach \$10.5 trillion annually by 2025¹
- **Data Breaches:** 22 billion records exposed in 2022²

❑ Increasing Attack Vectors

- **IoT Devices:** Billions of devices are vulnerable to attack due to weak security
- **Remote work:** Expand attack surface with unsecured home networks

1. <https://cybersecurityventures.com/cybercrime-damages-6-trillion-by-2021/>

2. <https://www.statista.com/topics/11610/data-breaches-worldwide/>

The Role of a Network Security Expert

❑ What do they do?

- **Detect Threats:** Monitor Traffic to identify anomalies
- **Prevent Attacks:** Implement firewalls, IDS/IPS, and secure configurations
- **Responds to Incidents:** Contains breaches, recover systems, and analyze vulnerabilities
- **Educate and Enforce:** Train users on secure practice and enforce security protocols

❑ Why it matters?

- **Prevent Catastrophic Losses:** Both financial and reputational
- **Enhance Organization Resilience:** Ensure business continuity

CS6903 Objective

- ❑ A solid foundation of network security concepts
- ❑ To learn how to monitor and analysis network traffic and protocols for detecting various types of attack/vulnerabilities
- ❑ Understanding how things work/break and how to fix them
- ❑ How to think like attacker or security expert?

Prerequisites

- ❑ Computer Networks
- ❑ Familiarity with Linux environment
- ❑ Strong programming skills (preferably C/C++/Python)
- ❑ Socket programming, Shell scripting

Administration

- ❑ Course Management through Moodle platform hosted at IITH;
 - First register on <https://moodle.cse.iith.ac.in>
 - Once registered, we will add you to the Course CS6903

- ❑ Teaching Slot:
 - D slot: Mon @ 12, Tue @ 8:30/9, and Fir @ 11
 - Venue: CS-LH2 (May change later depending on strength)

Tentative Grading Policy

- ❑ Class participation: 5%
- ❑ Quizzes: 10%
- ❑ Mid Sem Exam: 10%
- ❑ End Sem Exam: 15%
- ❑ Assignment/Projects: 60%
 - Programming Assignment/CTF/VAPT
 - Projects

Assignment/Project Policy

- ❑ Individual or Group based
 - Exact detail will be announced for each assignment/project
- ❑ Complete your assignment by your own
 - Can discuss with other, but not take solution from anyone else, including internet, ChatGPT, etc.
 - Please cite every third-party reference, if used any
- ❑ Group Assignment:
 - One group member can submit
 - Document each member's work in report or README file. Details will be shared with assignment document

Collaboration and Seeking Help

❑ Communicate with (other) Group members

- Clarifying problem, syntax doubt, debugging strategy
- Do not copy-paste from other group or Internet for any assignment
- Document each member's work including any third-party reference

❑ Engage with Instructors/TAs

- Discuss the problem being faced
- Explain your methodology
- Explain each member's responsibilities

Academic Integrity and Honesty

- ❑ Submitted work should be your own
- ❑ We have a **plagiarism** [policy](#) in place
- ❑ Any deviation will lead to “Fail (F)” grade
 - Case reported to disciplinary action committee

Ethics

- ❑ In this course, you will learn how to attack computer networking systems in a sandbox environment
- ❑ We learn attacks because it is needed to understand how to defend them!!
- ❑ You have an obligation to use this knowledge ethically
 - You **do not attack** others!
 - In addition to unethical, it may be a **crime**
 - Many good legitimate hacking challenges
 - <http://overthewire.org/wargames/> (wargames)
 - <https://challenges.re/> (reverse engineering challenges)
 - <https://ctftime.org/ctfs> (Capture the Flag competitions)

Reference Books/Materials

- ❑ ***Network Security: Private Communication in a Public World***, Kaufman, Perlman, and Speciner. Second Edition, Pearson, 2016
- ❑ ***Cryptography and Network Security***, William Stallings, Pearson, 7th Edition, 2016
- ❑ ***Security Engineering***, Ross Anderson, 2nd Edition ([free online](#))
- ❑ ***Computer Networking: A Top Down Approach***, James Kurose & Keith Ross, 7th Edition, Pearson, 2016.
- ❑ ***Attacking Network Protocols***, James Forshaw, 2017
- ❑ **Course on Moodle** → articles, videos, news, etc.